

HASAN TATAR

Full-Stack Engineer · Applied AI & LLM Systems

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SUMMARY

Software engineer building software since 2018, now shipping production systems for paying clients as a freelancer across Python/FastAPI and TypeScript/Next.js, with applied-AI work running in production. First author on an IEEE SIU 2026 publication analyzing Turkish-language performance degradation in a vision-language-action model. Completing a B.Sc. in Computer Engineering, expected 2027. Open to full-time roles, contract work, and freelance engagements in Istanbul or remote.

EXPERIENCE

Freelance Software & AI Engineer

Jan 2026 – Present

Independent contractor, Istanbul (remote). Parallel client engagements listed below.

US fleet-operations company · NDA client · bus fleet operations platform

May 2026 – Sep 2026

- Designed a two-plane architecture separating high-frequency location telemetry (HTTP API, Lambda, DynamoDB) from consistency-critical CRUD and live subscriptions (AWS AppSync GraphQL), across 5 CDK stacks and 43 resolvers.
- Encoded regulatory hours-of-service limits as a unit-tested pure module, including multi-driver crew math where a two-driver crew is legal on combined remaining hours neither driver could cover alone.
- Found four classes of silent AppSync resolver failure invisible to TypeScript, lint, and CDK synth by driving the platform's own evaluate-code API, then wired automated validation of all 48 resolvers into CI.

Apparel e-commerce brand · NDA client · AI customer service and ad automation

May 2026 – Aug 2026

- Built the orchestration layer of an async AI customer-service pipeline (FastAPI, async SQLAlchemy, PostgreSQL) ingesting HMAC-verified webhooks from four messaging channels through a ten-stage turn: security filter, intent classification, order lookup, model reply, output validation, send, persist, escalate.
- Cut model spend by short-circuiting intent classification with two zero-token heuristics ahead of any model call; roughly 67% of conversations were classified without one.
- Restructured the model request into cached and per-turn blocks after finding the previously fused system prompt broke provider-side prompt-cache prefix matching on every call.
- Eliminated duplicate customer messages during rolling deploys with a Postgres transaction-scoped advisory lock keyed on conversation id, layered over the existing per-process lock. Covered by one of 771 test functions across 43 modules over 19,094 lines of production Python.

GIS Analytics · freelancer network, London, UK · client projects routed through the network

Mar 2026 – Present

- Built web interfaces with interactive 3D rendering for geospatial data visualization, and full-stack features across the network's product from data layer to client.
- Built an industrial electrical distributor's website and migrated its catalog off a legacy Laravel CMS: an ETL parsing 2,798 bilingual records across 266 categories into 1,875 deduplicated products, with shared canonical and hreflang metadata and a locale-aware sitemap across three locales.
- Provided ongoing technical support to a ready-to-wear fashion brand, including Meta Ads account analysis and a scoped service proposal.

okeep.co / yaptir.io · freelance client · software and automation agency

Jan 2026 – May 2026

- Built the agency's marketing site (Vite, vanilla JS) with a custom three.js WebGL hero and a Groq-backed chat assistant; led a small engineering team and ran client-facing technical discussions across parallel engagements.
- Shipped automation workflows for AI-driven client operations, and closed the engagement with a security pass removing hardcoded credentials from the repository.
- Alongside client work, built Camellayer (a Turkish-to-English prompt compression engine iterated across 17 versions against measured tokenizer counts, with hooks for four AI coding CLIs) and Retro Lokal (a vintage storefront on Next.js 16, Supabase, and iyzico with a CraftJS page builder and a Flutter companion app).

Research & Development Intern Aug 2025 – Dec 2025 · Software Intern Jun 2025 – Aug 2025

- Built an image-based lost-baggage matching pipeline combining color, shape, and texture similarity with multi-angle, centroid-weighted voting to re-identify luggage across camera angles; sole author across 161 commits and 29 versioned releases.
- Integrated ESP, AD, and ADS hardware modules with the imaging software for global airline companies.

Earlier

Jul 2024 – Mar 2026

- **B4AFC, Business for Aging Friendly Community**, IT Specialist (part-time), Jan–Mar 2026: designed, built, and maintained the organization's website and internal digital tooling.
- **Independent**, Game Developer (freelance), Aug–Dec 2024: shipped an isometric RPG and a 2D office game in Unity and C#, handling gameplay design and art direction end to end.
- **NOVENTIQ Makronet Bilgi Teknolojileri**, Software Intern, Jul–Aug 2024: contributed to internal .NET and web tools; supported the data analytics team on reporting workflows.

SELECTED PROJECTS

Quantum: Windows-on-Apple-Silicon compatibility layer *Rust*

A translation stack written from scratch with no third-party crates: PE/COFF loader, x86-64 decoder, AArch64 emitter validated against clang-assembled golden output, a register-pinned JIT lifter, and a DXBC shader parser emitting Metal Shading Language into a D3D11-to-Metal device layer.

xtask: regulatory compliance platform for German PV installers *live demo* · wazder.github.io/powerbird

Fills the official 23-page VDE-AR-N 4105 AcroForm (557 fields across four grid-connection forms) from project data, verified by automated field assertions, and exports well-formed XRechnung 3.0 UBL invoices with the § 12 UStG VAT split applied. Keeping extracted sub-documents interactive required rebuilding the AcroForm through reference-based widget copying around a pdf-lib limitation.

Pull: real-time multiplayer typing game *Godot 4, Cloudflare Workers*

Server-authoritative 1v1 protocol across three Durable Objects: match room, matchmaking queue, SQL-backed leaderboard. The server computes rope position from its own cumulative per-player state rather than trusting client-reported positions, caps per-message movement, and rate-limits sabotage abilities.

PUBLICATION

H. Tatar, U. Gökmen, I. Akkoyun, and S. Kahraman, “Performance Analysis of Vision-Language-Action Models: Evaluating Turkish Command Understanding in OpenVLA,” in *Proc. 2026 34th Signal Processing and Communications Applications Conf. (SIU)*, İstanbul, Türkiye, Jul. 2026, pp. 1–4, doi: [10.1109/SIU71813.2026.11636432](https://doi.org/10.1109/SIU71813.2026.11636432).

First and corresponding author. 50 manipulation commands in matched Turkish/English pairs over 300 OpenVLA-7B inferences: mean L2 action error 0.501, cosine similarity 0.372 ($p < 0.001$), complexity uncorrelated to degradation, and 30.8% more tokens for Turkish after BPE segmentation.

EDUCATION

İstanbul Ticaret Üniversitesi · B.Sc. Computer Engineering

Sep 2023 – expected 2027

Fatih Sultan Mehmet Vakıf Üniversitesi · B.Sc. Computer Engineering, transferred

Sep 2022 – Jul 2023

SKILLS & CERTIFICATIONS

LANGUAGES	Python · TypeScript · JavaScript · Rust · Java · C# · C · Swift · SQL
BACKEND	FastAPI · Node.js · Spring Boot · .NET · PostgreSQL · SQLAlchemy · Supabase · REST APIs
FRONTEND	React · Next.js · Astro · Tailwind CSS · three.js
AI	PyTorch · Transformers · Anthropic Claude API · Computer Vision (YOLO/SAM inference and tracking)
INFRA	AWS CDK · Google Cloud · Cloudflare Workers · Vercel · Firebase · Docker · Git · CI/CD
SPOKEN	Turkish (native) · English (professional)
CERTIFICATIONS	Google Cloud Professional Cloud Architect · Anthropic Claude Developer track (Claude Code, Claude API, Agent Skills, MCP) · Yapay Zeka ve Teknoloji Akademisi Advanced AI program